

Retrieval-Augmented Generation for Communicable Disease Query Answering in Childcare Settings

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Abstract— There is a risk of misinterpretation and delayed response because childcare providers frequently do not have fast, dependable access to information about communicable diseases. In order to address disease-related questions based on the NH Disease Handbook for Childcare Providers, this paper presents a Retrieval-Augmented Generation (RAG) system that is powered by the Qwen3-8B large language model and PubMedBERT biomedical embeddings. To ensure traceable, handbook-grounded responses, sentence-aware chunking, FAISS vector indexing, and scope control are employed in this machine project. Qwen3-8B received the highest Correctness score of 0.686 among the three models. Hence, this model is a good fit for this domain. The results show the effectiveness of domain-specific retrieval grounding and biomedical embeddings in minimizing the hallucination risk, providing a useful reference tool for non-clinical childcare workers without replacing medical professionals.

Keywords— Retrieval-Augmented Generation, LLM, diseases, PubMedBERT, FAISS, NLP, public health, medical question answering

I. INTRODUCTION

Childcare providers face a daily challenge: quickly identifying whether a sick child's symptoms match a communicable disease, understanding how that disease spreads, and knowing when the child should be excluded from the facility. Misinformation or delayed access to accurate disease information can lead to outbreaks that affect children and staff. Accurate and timely access to medical information is still crucial for managing health and maintaining safe childcare settings. Childcare providers, many without medical training, often have to read long PDFs or handbooks to find symptom matches and exclusion rules. This process can take a lot of time and is easy to misinterpret in stressful situations. Recently, Large Language Models (LLMs) have introduced natural-language tools for asking medical questions more intuitively. Stand-alone LLMs in healthcare still have limitations. They may provide false information, use out-of-date data, and fail to disclose the source of their responses [17]. Models that are not properly grounded may produce advice beyond their intended scope, creating potential safety concerns in childcare settings.

Retrieval-Augmented Generation (RAG) architectures have been proposed to overcome these constraints by using data from external sources [5]. Large language models use real-time retrieval to obtain relevant information from knowledge stores [10]. Combining retrieval and generation retains outputs based on validated sources, reducing errors while maintaining accuracy and traceability [4]. Research shows that RAG systems can help

with clinical decisions, sharing patient information, and answering medical questions with clear sources [14] [18]. However, research on RAG systems for non-experts in childcare and pediatric settings is limited.

This paper proposes an automated medical information retrieval system based on a RAG architecture and powered by the Qwen3-8b model for treating communicable diseases in children. The system is designed as a reliable, always-available reference tool for childcare providers. Its knowledge base is established in the NH Disease Handbook for Childcare Providers, covering more than 40 communicable diseases. The solution combines sentence-aware chunking, biomedical domain-specific embeddings (PubMedBERT) [6], and an FAISS vector index to allow exact context retrieval and traceable natural language answers. The system strictly enforces scope control, refusing to answer questions outside the handbook.

RAG-based medical inquiry systems improve information access and reliability. They enable speedy assessment of symptoms and exclusion rules, reducing the need for manual searches, and promoting safe disease management. By using only validated sources, the approach ensures that providers receive accurate public health information rather than speculation.

The study's objectives are: (1) to investigate a data chunking and embedding strategy that captures disease-specific context without cross-contamination, (2) to develop and evaluate a medical RAG chatbot using the Qwen3-8b model, and (3) to analyze retrieval and generation performance using embedding metrics and LLM-as-a-Judge methodologies.

II. REVIEW OF RELATED STUDIES

A. Retrieval Augmented Generation (RAG)

In knowledge-intensive fields like healthcare, Retrieval-Augmented Generation (RAG) has become a promising paradigm for enhancing the dependability and factual foundation of large language models. When used in specialized decision-support environments, traditional large language models exhibit strong linguistic fluency but are susceptible to hallucinations, out-of-date knowledge, and unsupported inference [5], [10]. These limitations present serious safety risks in medical and public health applications because inaccurate or speculative outputs may result in incorrect guidance or misinterpretation of clinical data [17]. As a result, hybrid retrieval-generation frameworks that limit model outputs to validated knowledge sources have become more and more popular in research.

RAG-based architecture has been extensively investigated in healthcare natural language processing to facilitate clinical question answering, medical summarization, and patient education systems [1]. Source traceability, contextual relevance, and answer correctness are all significantly improved by the addition of external knowledge retrieval mechanisms [2] [4] [14]. Medical RAG systems are particularly helpful in situations where domain accuracy is more important than generative creativity, such as symptom triage, disease management guidance, and the distribution of public health information [13] [15]. By restricting response generation to retrieved evidence passages, RAG frameworks reduce the risk of hallucinations while retaining the flexibility of natural language interaction [18].

In biomedical NLP applications, domain-specific pretraining has also been demonstrated to improve retrieval quality. Compared to general-purpose language encoders, biomedical embedding models trained in scientific literature and clinical corpora offer semantic representations that more accurately capture medical terminology and symptom-disease relationships [6]. By enhancing semantic matching between user queries and knowledge base passages, dense retrieval techniques employing specialized embeddings perform better than sparse lexical retrieval techniques in medical question-answering tasks [9].

B. Model Architecture

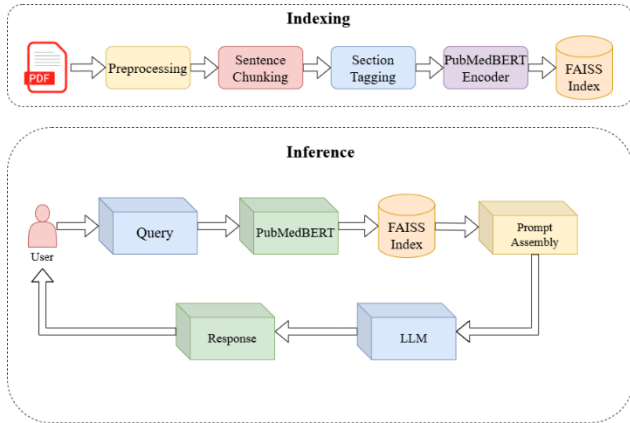


Fig. 1. RAG Architecture

As seen above in Figure 1, the suggested system utilizes a multi-stage RAG pipeline that includes vector indexing, biomedical embedding encoding, sentence-aware chunking, preprocessing, relevance validation, and generative response synthesis. The knowledge base was developed using the NH Disease Handbook for Childcare Providers, which is a source of standardized recommendations on the identification of communicable diseases, the modes of transmission, and exclusion criteria. The system was developed with stringent scope limits to ensure that the output is kept within the knowledge base, considering the fact that the users may not be medically trained.

1) Sentence chunking segmentation

The effectiveness of the retrieval performance in RAG systems is highly dependent on the segmentation of the documents before they are embedded. In this regard, the

use of sentence-level chunking has been employed in the proposed framework to ensure semantic coherence, minimal context dilution, and cross-topic contamination. Unlike the use of large chunk sizes, which might contain a mixture of different disease descriptions, the use of sentence-level segmentation ensures that the context is disease-specific, thus enhancing retrieval precision [8].

2) Biomedical Embedding with PubMedBERT

The encoding section of the proposed architecture incorporates PubMedBERT-style biomedical embeddings to represent text chunks in a high-dimensional space. The domain pretraining of the model, in particular, allows it to better capture clinical terminology, disease names, descriptors of disease symptoms, and epidemiological relationships compared to a general-purpose transformer. Domain-specific encoding has been demonstrated to enhance the accuracy of downstream retrieval tasks, including contextual alignment in medical NLP applications [6]. The use of a similar space to represent queries and handbook texts enhances the robustness of similarity matching in question answering systems. This was further supported by Li et al., [11] where the PubMedBERT model achieved better performance than the general transformer models (e.g., BioBERT, BlueBERT, and SciBERT). That achieved F1-scores of 0.715, 0.836, and 0.622 respectively from three (3) existing clinical trial eligibility corpora in their study. This supports that adding more training data from other related datasets attribute to the improving named-entity recognition (NER) accuracy. By the idea that cross-corpus data augmentation is beneficial for performance improvement.

3) Vector Retrieval with FAISS Indexing

The encoded embeddings are stored and searched using FAISS vector indexing to facilitate efficient similarity searches. FAISS is a popular vector search library used in many large-scale retrieval systems due to its optimized approximate nearest neighbor search algorithms that facilitate real-time query processing even in the presence of a large number of passages in the knowledge base [3] [9]. In the proposed system, FAISS acts as the core retrieval system that retrieves the top-ranked disease-related contexts based on cosine similarity scores.

4) Qwen3-8b

Qwen3-8B is used in the architecture's last stage to synthesize natural language responses. Its capacity for prompt following, support for various languages, and strong reasoning ability qualify it as an instruction-tuned large language model with good results for using it with RAG pipelines [10]. Qwen3-8B requires fewer epochs during training with better classification accuracy compared to other conventional transformers like BERT, RoBERTa, and T5, according to recent studies [10]. The optimized techniques for fine-tuning improve efficiency in using this model for specific NLP operations.

In line with previous findings that RAG enhances factual accuracy over standalone LLMs [11], Qwen3-8B is incorporated into the retrieval-augmented generation framework in this study, where retrieved handbook passages are combined with user queries to produce grounded and coherent responses.

III. METHODOLOGY

This study employs an experimental research design to evaluate the effectiveness of a Retrieval-Augmented Generation (RAG) pipeline for answering communicable disease queries grounded in the NH Disease Handbook for Childcare Providers, published by the New Hampshire Department of Health and Human Services (DHHS) [19]. The core generative model is Qwen3-8B, a state-of-the-art open-source large language model (LLM), deployed under 4-bit NF4 quantization to enable experimentation within the memory constraints of a Google Colab T4 GPU environment.

The methodology is organized into five phases: (1) knowledge base preparation and preprocessing, (2) RAG pipeline construction, (3) retrieval and generation evaluation, (4) LLM as a Judge quality assessment, and (5) hallucination and abstention analysis. The research follows a controlled experimental approach in which all evaluation queries are assessed against an identical, frozen retrieval layer — ensuring that observed differences in answer quality are attributable solely to the generative model's behavior given the same retrieved context.

A. Dataset Preparation

1) Knowledge Source

The knowledge base consists of a single publicly available PDF document the NH Disease Handbook for Childcare Providers obtained from the official New Hampshire DHHS Division of Public Health Services [19]. The handbook is a comprehensive reference covering over 40 communicable diseases commonly encountered in childcare settings and is legally in the public domain, collected for research and educational purposes.

TABLE I.
DATASET STRUCTURE

Property	Value
File Type	PDF (.pdf)
Number of source documents	1
Estimated pages	~180 pages
Diseases covered	~40+
Approximate text size	~120,000 characters

2) Dataset Characteristics

The handbook is organized as a series of structured disease fact sheets, each covering a single disease with standardized subsections: What is it? / What are the symptoms? / How is it spread? / How soon do symptoms appear? / Who is at risk? / How is it treated? / Can it be prevented? / Exclusion criteria / Is it reportable? These structural properties provide semantic regularity that supports retrieval while introducing challenges from recurring headers, footers, and page numbers embedded across all pages during PDF rendering.

B. Data Preprocessing

Raw PDF content undergoes several preprocessing steps before being used as the knowledge base for the RAG pipeline. The preparation pipeline consists of three stages: text extraction, text cleaning and normalization, and

document chunking with metadata creation. All stages are written in Python.

1) PDF Text Extraction

Text extraction is carried out using the PyPDF2 package, which reads the PDF document page by page and delivers the raw text content as Python strings. This page-by-page approach preserves the relationship between extracted content and its original page number. The extractor goes over all pages, saves the extracted text with the page number, and skips pages that have no extractable text.

2) Text Cleaning

The cleaning pipeline applies the following transformations to each extracted page:

- Hyphen line break repair: Reconnects words erroneously split across lines by PDF conversion artifacts (e.g., converting "dis-ease" to "disease").
- Header and footer removal: A suite of over 20 regular expression patterns is deployed to systematically excise recurrent non-content elements such as institutional headers, telephone numbers, physical addresses, URLs, and revision notes that appear consistently across the document's pages.
- Page number removal: Standalone numeric page numbers are stripped.
- Whitespace normalization condenses superfluous whitespace and lowers several line breaks to a single, uniform space.
- Spaced word repair: Employs targeted regular expressions to correct erroneous intra-word spaces caused by formatting discrepancies.

C. Document Chunking

- Chunk size (5 sentences): Provides enough context for PubMedBERT embeddings to encode significant medical concepts while being specific enough for accurate retrieval. This is consistent with Lewis et al. [10], who suggest passage-level granularity of about 4-6 phrases for maximum open-domain QA retrieval performance.
- Overlap (1 sentence): Maintains semantic continuity across structural boundaries, ensuring the retrievability of contextual information spread over distinct chunks. This technique reduces the likelihood of factual fragmentation without adding unnecessary redundancy.
- Disease-aware sectioning: In addition to fixed-size chunking, the implementation detects disease section boundaries using a keyword dictionary and labels each chunk with the appropriate disease section. This maintains topical coherence inside chunks while also preventing cross-disease contamination, which is crucial for a medical reference because disease-specific information should not be jumbled across sections.

The resulting corpus contains 482 chunks from the disease handbook, averaging approximately 5 sentences per chunk across 22 disease sections.

D. Retrieval-Augmented Generation (RAG) Pipeline

This study's system architecture is centered on the RAG pipeline. It consists of two major components: an offline indexing phase that creates and stores all retrieval artifacts, and an online inference phase that uses those artifacts to answer user queries. This split ensures that the generative model always runs against the same, frozen retrieval layer.

1) Embedding Model

All text chunks are represented as dense vectors by the Sentence Transformers library's S-PubMedBert-MS-MARCO model (pritamdeka/S-PubMedBert-MS-MARCO). This pre-trained biomedical encoder converts variable-length text sequences into fixed-dimensional dense vectors, capturing domain-specific semantic meaning.

TABLE II.

EMBEDDING PROCESS

Property	Value
Embedding Model	pritamdeka/S-PubMedBert-MS-MARCO
Model Type	Sentence-BERT fine-tuned on PubMed + MS MARCO
Embedding Dimension	768
Chunk Size	5 Sentences
Overlap	1 Sentence
Normalization	L2 Norm (For cosine similarity)

The same embedding model is used at both indexing time and query time, ensuring that chunk and query vectors occupy the same vector space.

E. Vector Indexing

All chunk embeddings are indexed using FAISS (Facebook AI Similarity Search) with a flat inner-product index (IndexFlatIP) over the embedding space. Prior to indexing, all vectors are L2-normalized using faiss.normalize_L2, making inner-product search equivalent to cosine similarity search. The flat index performs exhaustive exact nearest-neighbor search without approximation, which is feasible given the corpus size of 482 vectors.

During inference, a natural language query is encoded with the same Sentence Transformer model to produce a query vector, which is also L2-normalized before search. FAISS returns the top-k = 7 closest chunk vectors based on inner-product distance, which corresponds to the most semantically similar passages from the handbook.

F. Generative Model

The generative backbone of the system is Qwen3-8B, loaded from Hugging Face using the transformers library with 4-bit NF4 quantization via BitsAndBytesConfig. Quantization reduces the model's GPU memory footprint from approximately 32 GB to approximately 5 GB, enabling deployment on a Google Colab T4 GPU (15 GB

VRAM). The model is loaded with bfloat16 compute dtype and device_map="auto" for automatic layer placement.

G. Prompt Construction and Context Injection

The retrieved pieces are concatenated to produce a structured context block, which is then incorporated into a rigorous, extraction-focused prompt. This setup constrains the generative model to rely entirely on the available evidence. Furthermore, the prompt uses the Qwen3 architecture's native chat template, with enable_thinking=False specifically set to disable chain-of-thought reasoning processes, resulting in short and deterministic responses.

Generation parameters are configured as follows to maximize factual precision and minimize hallucination:

- max_new_tokens: 300
- do_sample: False (greedy decoding)
- repetition_penalty: 1.15

H. System Evaluation

This section evaluates the RAG pipeline on a held-out set of 43 questions that were not used during development. The evaluation covers five difficulty tiers: easy, medium, hard, and natural language queries.

1) Evaluation Setup

Evaluation was performed using manual benchmark questions designed to simulate realistic childcare provider queries. The 43 questions span:

TABLE III.

QUESTION TYPES

Tier	Count	Description
Easy	11	Single-fact, single-disease questions
Medium	12	Multi-step or procedural questions
Hard	10	Cross-disease comparisons, policy questions
Natural	10	Informal, colloquial queries

Each question categories presented above in Table III are generated from Claude AI Sonnet 4.6.

TABLE IV.

SAMPLE QUESTIONS

Question Type	Query
Easy	"What is the incubation period of chickenpox?"
Medium	"What steps should be taken if a child is diagnosed with head lice?"
Hard	"What are the immunization requirements for children attending childcare in New Hampshire?"
Natural	"My kid has a rash and a fever what could it be?"

Table IV above presents sample queries from Claude AI Sonnet 4.6. Where each question are annotated with ground-truth keywords (used to label retrieved chunks as relevant) and an expected answer.

IV. RESULTS AND ANALYSIS

This section presents the quantitative evaluation of the proposed Retrieval-Augmented Generation (RAG) system integrating Qwen3-8b as the generative model and PubMedBERT for embedding-based retrieval.

The system was evaluated using 43 questions generated via Claude, categorized into five difficulty levels: easy, medium, hard, and natural. Retrieval depth was fixed at $k=7$. Performance was measured using both retrieval metrics (Recall@7, Precision@7, Context Relevancy) and LLM-based evaluation metrics (Faithfulness, Answer Relevancy, Correctness).

A. Overall Performance (Qwen3-8B)

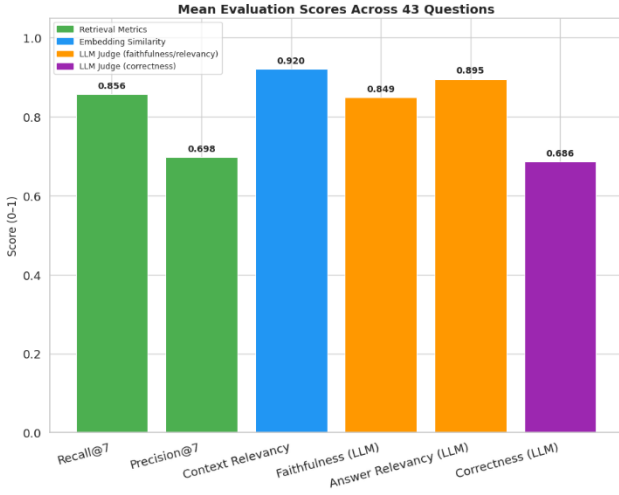


Fig. 2. Mean Evaluation (43 Questions)

Figure 2 above shows the mean performance of the Qwen3-8b model when presented 43 Questions.

TABLE V.
OVERALL PERFORMANCE ACROSS IN-SCOPE

Metric	Score
Recall@7	0.856
Precision@7	0.698
Context Relevancy	0.920
Faithfulness (LLM)	0.849
Answer Relevancy (LLM)	0.895
Correctness (LLM)	0.696

1) Retrieval Effectiveness

- Recall@7 = 0.856
 - Indicates that most relevant documents are successfully retrieved within the top 7 passages.
- Precision@7 = 0.698
 - During the retrieval phase, approximately 70% of the documents obtained were relevant. This shows that the system is set up to cast a wide net and capture as much context as possible, rather than strictly filtering for exact matches.

- Context Relevancy = 0.920
 - The retrieved passages and the query have a very high level of semantic alignment. This confirms that PubMedBERT embeddings successfully represent biomedical semantics.
- 2) *Generation Quality*
 - Faithfulness = 0.791
 - While the model generally bases its answers on the available context, we did notice several instances where it extended beyond the source information or incorporated slight mistakes.
 - Answer Relevancy = 0.919
 - The generated answers closely match the query purpose. Qwen3-8B makes great use of retrieved passages.
 - Correctness = 0.698
 - Suggests that while responses are relevant and mostly grounded, full biomedical accuracy is not always guaranteed.

B. Overall Performance Comparison

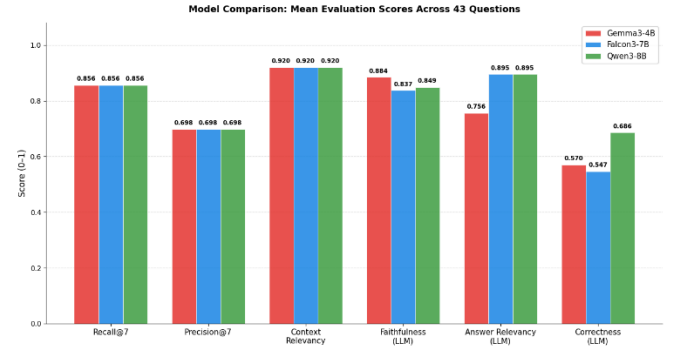


Fig. 3. Mean Evaluation across three LLMs (43 Questions)

Figure 3 above shows the comparison between the three models (Gemma3-4b, Falcon3-7b-instruct and Qwen3-8b) when presented 43 Questions.

TABLE VI.
MODEL COMPARISON ACROSS ALL METRICS

Metric	Gemma3-4B	Falcon3-7B-instruct	Qwen3-8B
Recall@7	0.856	0.856	0.856
Precision@7	0.698	0.698	0.698
Context Relevancy	0.920	0.920	0.920
Faithfulness	0.884	0.837	0.849
Answer Relevancy	0.756	0.895	0.895
Correctness	0.570	0.547	0.686

1) Retrieval Effectiveness

Retrieval metrics (Recall@7, Precision@7, Context Relevancy) remained identical across all models, confirming that performance differences are attributable solely to the generative component.

2) Generation Quality

- Gemma3-4b
 - achieved the highest Faithfulness score (0.884), suggesting it stays closest to the retrieved context when generating responses.

- Falcon3-7b-instruct
Falcon3-7B and Qwen3-8B tied on Answer Relevancy (0.895), indicating both models effectively interpret and address user queries.
- Qwen3-8b
achieved the highest Correctness score (0.686), outperforming Gemma3-4B (0.570) and Falcon3-7B (0.547) by a significant margin.

Overall, the best generation quality was demonstrated by Qwen3-8B, with the highest Correctness score, suggesting that the increased number of parameters helps provide the most accurate answer when grounded in the provided context. Interestingly, the disparity between the models is small, suggesting that models of lower capacities, such as Gemma3-4B and Falcon3-7B, remain viable within the RAG framework. This implies that retrieval grounding may mitigate the limitations of model capacity and that the RAG framework is robust enough to provide quality answers despite the model's limitations.

TABLE VII.
OPTIMAL NUMBER OF RETRIEVED CHUNKS

K	Recall@K	Precision@K	Context Relevancy
1	0.1855	0.8140	0.9312
3	0.4983	0.7287	0.9259
5	0.7643	0.7287	0.9223
7	0.8564	0.6977	0.9202
10	0.8880	0.6628	0.9174

To identify the optimal number of retrieved chunks (K), the system was tested at $K \in \{1, 3, 5, 7, 10\}$, where K represents the number of passages retrieved from the FAISS index per query. As shown in Table VII, Recall@K rises with K while Precision@K drops, and Context Relevancy remained stable across all values (0.917–0.931). The marginal recall gain from K=7 to K=10 (+0.032) did not justify the precision loss (−0.035), making K=7 the optimal balance between retrieval coverage and prompt quality.

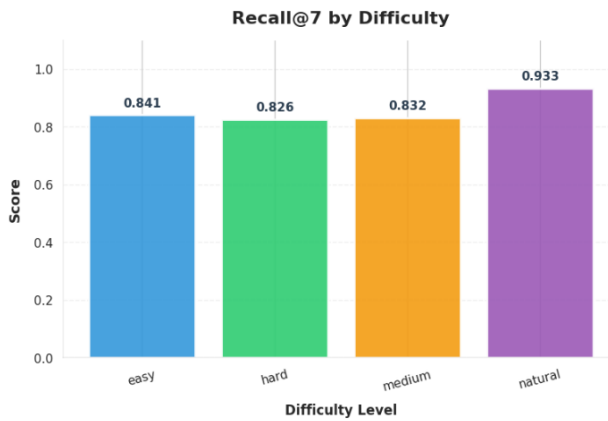


Fig. 4. Recall@7 for Difficulty

TABLE VIII.
RECALL PERFORMANCE ON QUESTIONS

Question Type	Score
Easy	0.841
Hard	0.826
Medium	0.8
Natural	0.933

1) Recall@7 by Difficulty Analysis

- Natural questions (0.933) had the highest recall. Most likely because training corpora use terminology that is more similar to real-world biological jargon. When inquiries are similar to natural clinical terms, embedding similarity improves.
- Easy, Medium, and Hard (0.826-0.841) have steady recall, demonstrating the robustness of PubMedBERT embeddings under organized difficulty fluctuations. As the intricacy of the questions rises, there is negligible performance decrease.

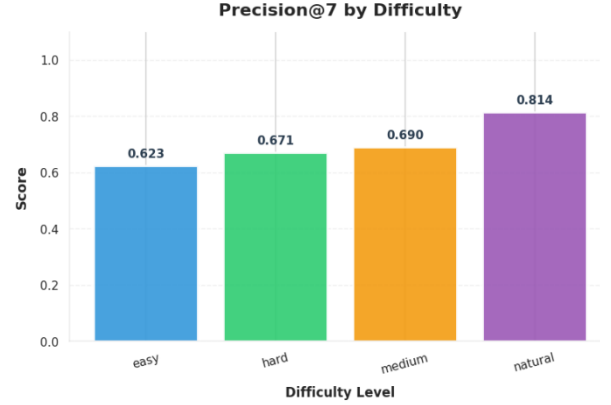


Fig. 5. Precision@7 for Difficulty

TABLE IX.
PRECISION PERFORMANCE ON QUESTIONS

Question Type	Score
Easy	0.623
Hard	0.671
Medium	0.690
Natural	0.814

2) Precision@7 by Difficulty Analysis

- The system performed best on Natural questions (0.814), indicating that organic phrasing aligns well with how biological concepts are clustered in the embedding space.
- Medium and Hard questions showed moderate precision (~0.67–0.69). Here, more complex queries seemed to retrieve larger, related concepts rather than specific, pinpointed evidence.
- Easy questions lagged at 0.623. The lower precision here likely stems from shorter queries

retrieving nearby context that isn't strictly necessary for a simple answer.

C. System Interpretation

The experimental results reveal three important characteristics of the proposed RAG architecture:

- **Retrieval Foundation:** PubMedBERT embeddings provide high recall and excellent contextual relevance, validating the choice of a domain-specific encoder.
- **Effective Context Utilization:** Qwen3-8b provides relevant and grounded answers, suggesting that retrieved portions steer generation.
- **Bottleneck:** The gap between Faithfulness (0.791) and Correctness (0.698) suggests that: retrieved context is relevant, but the generator does not always synthesize fully accurate biomedical conclusions.

V. DEPLOYMENT

The chatbot is deployed as an interactive web application using Gradio. The interface allows users to type natural language questions and receive answers grounded in the childcare disease handbook, with retrieved source chunks displayed in a side panel.

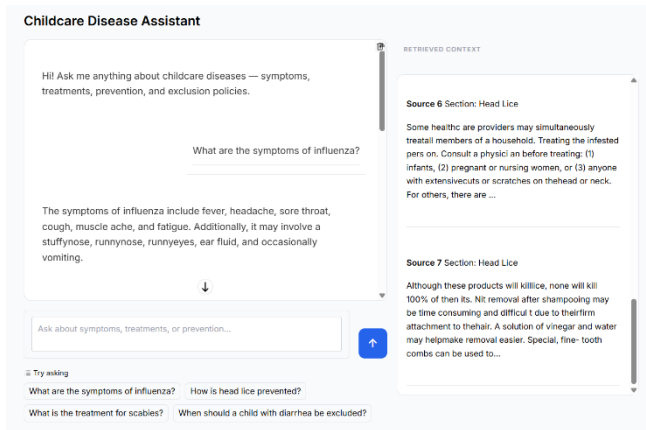


Fig. 6. Gradio Deployment

As observed above in Figure 5, the interface presents the following:

- Chat interface: Conversational Q&A with message history.
- Retrieved context panel: Displays the top-7 source chunks used to generate each answer.
- Example questions: Pre-loaded examples for quick exploration.
- Real-time generation: Responses generated live using Qwen3-8b.

The application was deployed via Gradio's public sharing feature (share=True), generating a public URL accessible from any device without local installation.

VI. DISCUSSION

The experimental evaluation of the proposed Retrieval-Augmented Generation architecture yields critical insights into the viability and limitations of deploying quantized open-source large language models for specialized public health applications.

A. Retrieval Robustness

The system demonstrated highly robust retrieval capabilities, a success primarily driven by the integration of PubMedBERT embeddings and the disease-aware sentence chunking strategy. The consistently high mean Recall@7 (0.856) and Context Relevancy (0.920) across different query types show that the encoding approach helped align everyday childcare questions with formal medical terms. The framework achieved its highest recall (0.933) and precision (0.814) on natural language questions. This highlights the system's resilience to the conversational phrasing typical of end-users in stressful childcare environments, showing that natural clinical expressions tightly cluster in the biomedical embedding space.

B. Generative Performance and Reasoning Bottlenecks

The retrieval module performed exceptionally well, but the generative tests exposed some reasoning limits common to smaller, 8B-parameter models. Qwen3-8B scored a solid 0.919 for Answer Relevancy, proving it can accurately interpret user intent and integrate the retrieved text. However, Faithfulness and Correctness decreased to 0.791 and 0.698, respectively, showing that retrieving the correct background information does not always ensure a fully accurate or hallucination free response. The model faced greater difficulty with complex queries, especially when it needed to compare symptoms and exclusion rules across different sections of the handbook. An 8B model is excellent at extracting and summarizing straightforward facts. However, its multi-step reasoning is less reliable, leading to minor inaccuracies when it has to synthesize information from multiple sources.

C. System Safeguards and Practical Implications

Despite the limitations seen in more complex queries, strict use of the NH Disease Handbook reduced the hallucination risks often linked to standalone LLMs. By grounding the Qwen3-8B model's responses only in retrieved content and applying clear semantic boundaries, the system favors verifiable information instead of open-ended generation. The findings indicate that trustworthiness in medical AI does not strictly require massive, proprietary models with hundreds of billions of parameters; rather, it can be achieved through rigorous domain-aware data segmentation and strict retrieval grounding. The deployment of this framework demonstrates that open-source RAG systems can safely serve as daily reference tools for public health administration, provided they act as information-lookup assistants rather than diagnostic replacements.

VII. CONCLUSION

This study presented an automated medical information retrieval system based on a Retrieval-Augmented

Generation architecture driven by the Qwen3-8B model and PubMedBERT embeddings, which is intended to serve as a reference tool for childcare experts. The results reveal that the system performed well in terms of getting relevant material and maintaining scope control, extracting appropriate context for natural language questions, and rejecting requests that fell outside of its coverage to avoid providing unverified medical advice. The use of tailored embeddings and organized sentence chunking helped keep responses focused on the correct topic. However, although the model demonstrated a clear understanding of user intent, complex queries involving multiple conditions were more difficult, leading to reasoning limitations and lower faithfulness scores typical of 8B parameter models. Despite these limitations, strict retrieval grounding and controlled knowledge boundaries reduced the hallucination risks associated with standalone language models. The deployment of this quantized open-source framework also demonstrates its feasibility for public health use as a daily reference tool without replacing clinical judgment. Future work will aim to improve generative precision and multi-step reasoning through hybrid search methods, explicit out-of-scope detection, and improved query decomposition to strengthen performance in real-world healthcare settings.

VIII. ACKNOWLEDGEMENT

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TABLE X.
QUESTIONS

Question No.	Query
1	"What is the incubation period of chickenpox?"
2	"What steps should be taken if a child is diagnosed with head lice?"
3	"What are the immunization requirements for children attending childcare in New Hampshire"
4	"My kid has a rash and a fever what could it be?"
5	"How should a childcare facility handle an outbreak of norovirus?"

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